Vote&Check: Secure Postal Voting with Reduced Trust Assumptions

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PETS 2025
17 July 2025











Why postal voting?

1. Postal voting is widely used
   - E-voting was suppose to replace postal voting (in practice: complement it).
   - Also for high-stage elections (eg. Tory leadership election in 2022)

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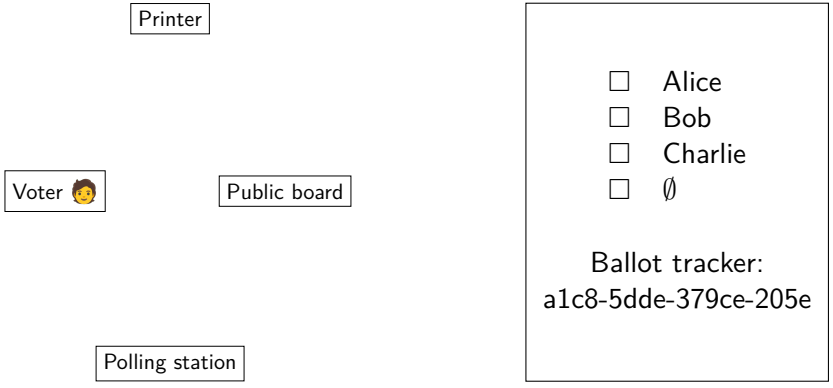
1. Postal voting is widely used
   - E-voting was suppose to replace postal voting (in practice: complement it).
   - Also for high-stage elections (eg. Tory leadership election in 2022)
2. Lack of understanding of security for postal voting
   - Only recent academic interest:
     - STROBE [Benaloh, E-Vote-ID'21]
     - RemoteVote & SAFE Vote [Crimmins, Rhea, Halderman, FC'22]
     - Louvain [Devillez, Pereira, Peters, 2024]
   - No systematic analysis, still strong trust assumptions

Contributions

1. Generic attacks on existing protocols
   - Breaking both privacy and verifiability
   - Set realistic objectives
2. New protocol: Vote&Check
   - Only simple cryptography
   - Reduced number of authorities
3. Formal proofs
   - New definitions
   - Proverif models

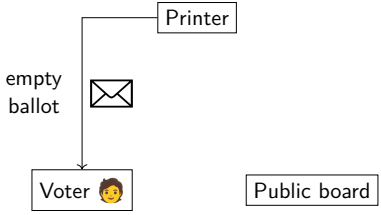
Generic protocol

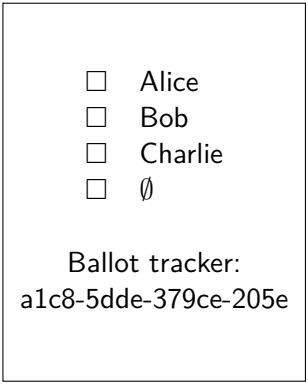
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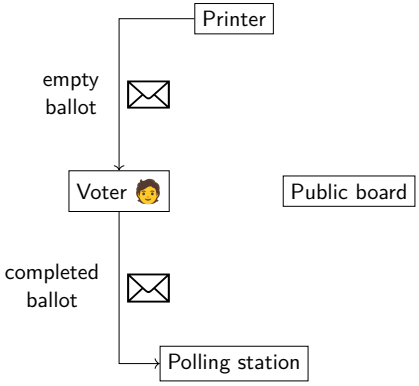
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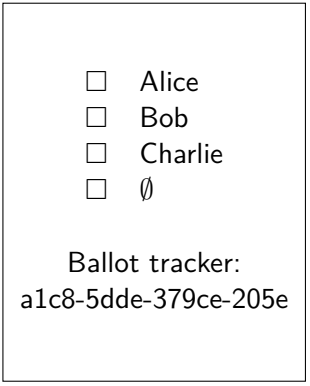




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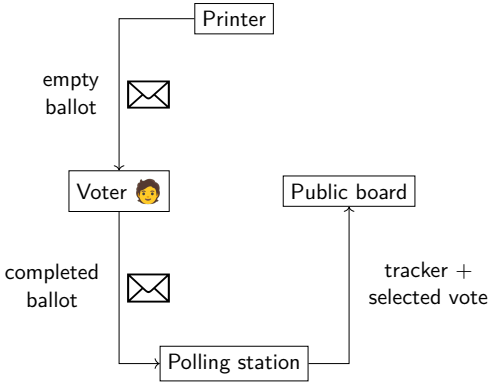
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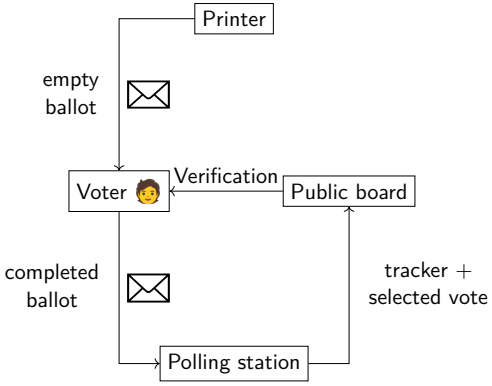


☐ Alice
☐ Bob
☐ Charlie
☐ ∅

Ballot tracker:
a1c8-5dde-379ce-205e

Generic protocol

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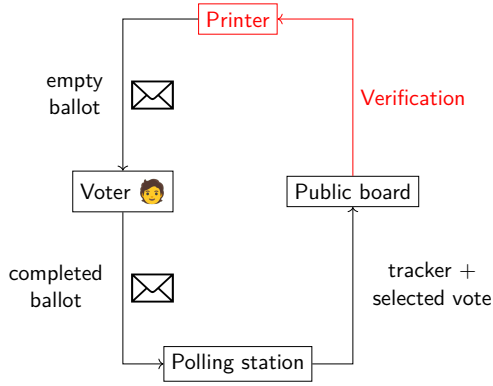
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Generic attack: full privacy breach

Attack from a dishonest printer

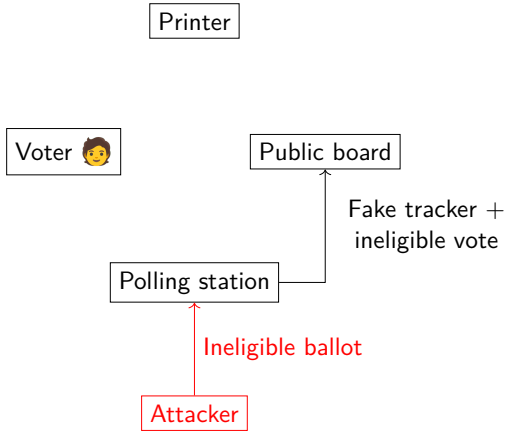
1. Printer emits material as usual, and keeps the ballot-voter link
2. Voters vote as usual
3. Printer performs verification instead of the voters



Generic attack: vote flooding

Attack from a dishonest third-party

- Attacker can forge ballots with arbitrary trackers
- Polling station does not know which trackers are valid
- If voters abstain, this will be unnoticed



Generic attacks

Protocols allowing voting without smartphone:

		STROBE	RemoteVote	SAFE Vote	Louvain	**Vote&Check**
	# of authorities	3	3	3	5	3
privacy attacks	full privacy breach					
	targeted privacy breach**new**					
	privacy breach by complaints**new**					
verifiability attacks	vote flooding**new**					
	clash attack**new**					
	alter non verifying voters					
	weak eligibility					
	universal verifiability				proxy	

new: attack in the author's threat model

- Remaining attacks seem inherent to postal voting
- Compared to Louvain:
  - less authorities
  - no need of a proxy verifier

New proposal: Vote&Check

Goal: design a verifiable postal voting protocol with security proofs

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Strong requirements:

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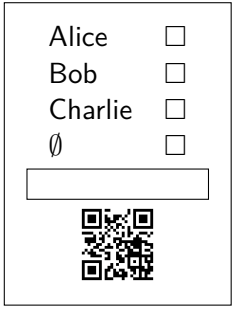
Good to have:

- Reduced number of authorities
- Only standard cryptographic primitives

POV: you are the voter

Step 1: receiving by post

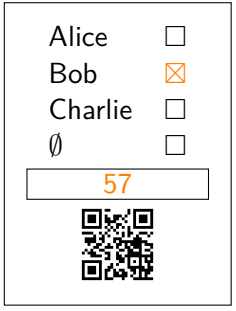




POV: you are the voter

Step 2: voting

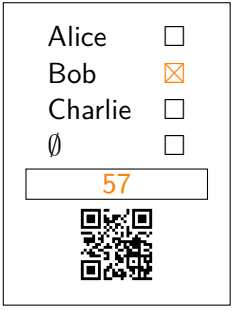




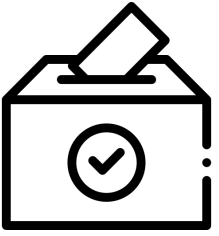
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Step 2: voting





Sent by the post

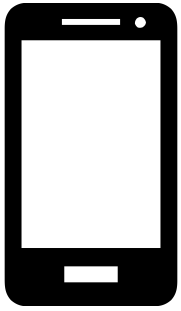


POV: you are the voter

Step 3: checking (optional)

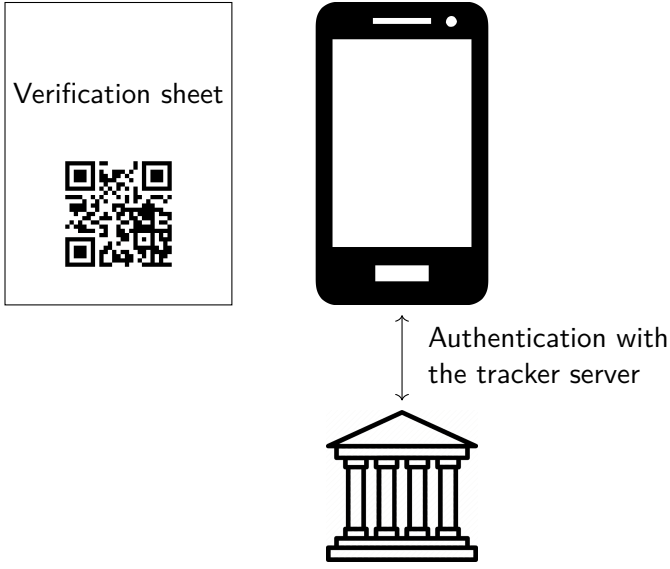
Verification sheet





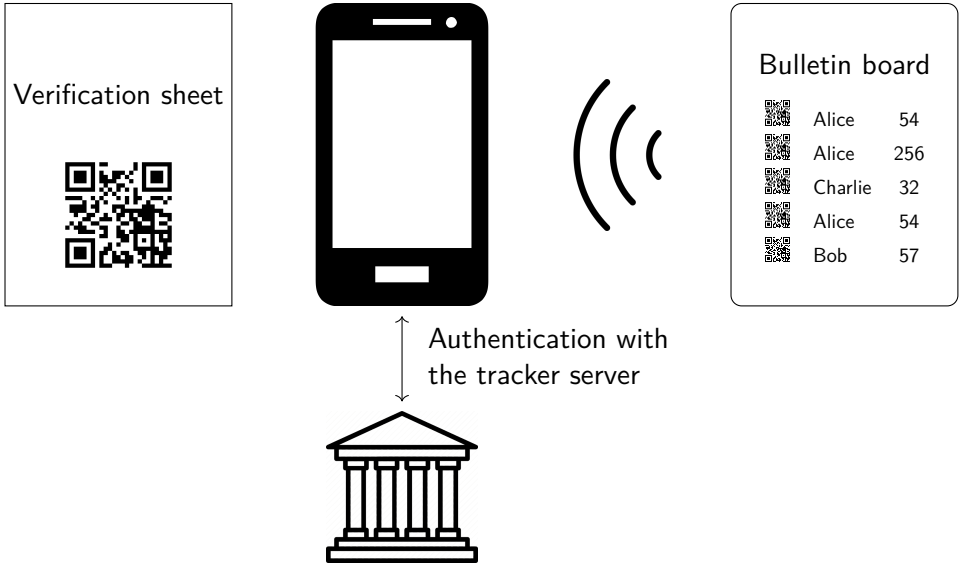
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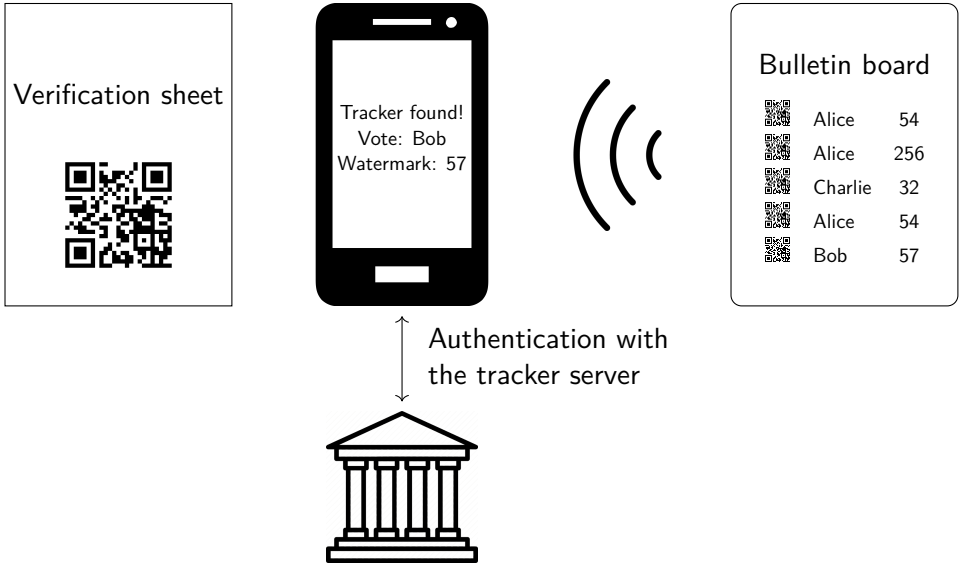
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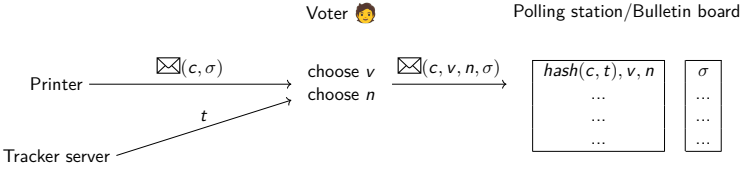


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Security mechanisms



Privacy through trust distribution

tracker $= \mathsf{hash}(c, t)$

- c is known only by the printer
- t is known only by the tracker server

No clash

A number chosen by the voter guaranties uniqueness on the bulletin board

No ballot forgery

All paper ballots are signed by the printer

Security claims



Formal proof using the Proverif automated protocol analyser

Tracker Server	Printer	Election Office	Privacy	E2E Verifiability	Weak verifiability	Individual verifiability
😇	😇	😇	✓	✓	✓	✓
😈	😇	😇	✓	✓	✓	✓
😇	😈	😇	✓	✗	✗	✓
😇	😇	😈	✓	✗	✓	✓
😇	😈	😈	✗	✗	✗	✓
😈	😇	😈	✗	✗	✓	✓
😈	😈	😇	✗	✗	✗	✓
😈	😈	😈	✗	✗	✗	✓

- Privacy holds if at most one authority is dishonest.
- Individual verifiability is unconditional.
- Dishonest election office can modify votes from lazy voters.
- Dishonest printer can cast ballots for abstainers.

Conclusion

New protocol: Vote&Check

- Brings privacy and verifiability to postal voting
- Only simple cryptography (signatures + hash functions)
- Reduced number of authorities
- Formal security proof in numerous scenarios

Limitations and future works

- Accountability
- Coercion resistance

Open question

- We said that some attacks seem inherent to postal voting (weak eligibility, alter non verifying voters).
- Prove it or find a solution?